

Service Certification for Agentic Wireless Networks: Extended Experiments and Code

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This research archive accompanies *Service Certification for Agentic Wireless Networks: Principles and Fundamental Limits*. We collect the complete experimental description from the supplied supplementary source, including additional numerical comparisons, figure data, and limitations. We retain each study's own population, comparator, and resource accounting.

Available material. The package supplies plotting scripts, summary data, editable figures, and protocol descriptions. The original simulation/controller drivers, complete trial records, and research directories cited in the source manuscript were not supplied. Accordingly, the scripts regenerate available figures; they do not rerun those experiments. Unmodified source fragments remain in `docs/source/`, and `docs/provenance.md` maps their contents. Statements about historical numerical outcomes below report the supplied manuscript's results, without asserting a new experimental replay.

Definitions for this standalone protocol

For the nine-pair wireless instance, R_i , f_j , and t_j denote radio rate, processor rate, and transport delay. The supplied manuscript sets $R = (20.574, 51.891, 100.556)$ Mbit/s, $f = (1, 2, 3)$ GHz, and $t = (0.8, 1.2, 1.8)$ ms. Tasks contain 80 kbit and require eight million processor cycles. With radio and server corrections r_i, w_j in ms, we use

$$\begin{aligned} D_{ij}(\theta) &= d_{ij}^0 + r_i + w_j, \quad d_{ij}^0 = 80/R_i + 8/f_j + t_j, \\ D_i^{\text{rad}}(\theta) &= 80/R_i + r_i. \end{aligned} \tag{1}$$

Charges satisfy $p_{ij} = p_i^{\text{rad}} + p_j^{\text{srv}}$, with $p^{\text{rad}} = (1, 2, 3.6)$ and $p^{\text{srv}} = (0.6, 1.4, 2.8)$. For goal parameters $(\alpha_g, T_g, T_g^{\text{rad}})$, the service decision is

$$\begin{aligned} \min_{i,j} \quad & p_{ij} + \alpha_g D_{ij}(\theta) \\ \text{subject to} \quad & D_{ij}(\theta) \leq T_g, \\ & D_i^{\text{rad}}(\theta) \leq T_g^{\text{rad}} \quad \text{when requested.} \end{aligned} \tag{2}$$

The acquisition studies set $\alpha_g = 0$. Symbol Φ denotes the standard normal cdf and $z_p = \Phi^{-1}(p)$. The following reference restates the scalar frontier used by the protocol; its proof remains in the paper and theoretical supplement.

Theorem 1 (Exact scalar completion reference). *For centered Gaussian probe means $a_e \vartheta$, let $q_e = a_e^2 / \sigma_e^2$ and let $A_{\max}$ be the greatest total precision available from history and a legal integer acquisition design. For $A_{\max} > 0$ and $0 < \delta < 1/2$, the attainable correct-completion probability at $\vartheta \neq 0$ is*

$$\Phi\left(|\vartheta| \sqrt{A_{\max}} - z_{1-\delta}\right).$$

A maximum-precision fixed design and its two-threshold endpoint rule attain this value. Completion $1 - \eta > \delta$ is possible exactly when $|\vartheta| \sqrt{A_{\max}} \geq z_{1-\delta} + z_{1-\eta}$.

At zero precision, operational policies abstain; the randomized reference can attain correct completion δ by assigning probability δ to each answer.

1 Experimental Protocols and Matched Comparisons

Every comparison retains its own population, verifier, and denominator. Costs include all attempts and acquired reports; additional costs exclude sunk history, and modeled durations exclude computation. The availability statement above applies to the upstream records named in the original source.

1.1 Wireless Model and Allocation

The nine-pair fixture uses evaluation-only correction means

$$(r_1, r_2, r_3) = (2.409418, 1.208761, .751280), \\ (w_1, w_2, w_3) = (6.061585, 2.113124, .935895).$$

in milliseconds. Rates use bandwidths (10, 15, 20) MHz and SNRs (5, 10, 15) dB. Aggregate and radio-counter directions are $a_{ij} = \mathbf{e}_i + \mathbf{e}_{3+j}$ and $\mathbf{e}_i$; response coefficients follow (1)–(2). Background radio/server means are $80/R_i$, $5/f_j$, and tagged processor service is $8/f_j$. Workload caps $\bar{w}_k$ equal seven background means. Reports average $m_{\text{coh}} = 64$ independent finite-FCFS-population draws, with proxy

$$\sigma_e^2 = (4m_{\text{coh}})^{-1} \sum_k a_{e,k}^2 \bar{w}_k^2.$$

This independence concerns synthetic diagnostics, not production packets. Aggregate costs/durations are $1 + .35p_{ij}$ and $1 + d_{ij}^0/20$; the R1 counter uses 12/2 units. The window is 20000, caps are $\bar{n}_e = h_e + \lfloor 20000/\ell_e \rfloor$, and no monetary cap binds separately.

Legacy scope. Missing runners, seeds, paired outcomes, and bootstrap records prevent reproducing the earlier stage/wireless comparisons. Adverse cases include: scalar quotas cost 14.79 versus residual’s 15.83 at 12.6 ms; loose-goal restart optimizer gaps exceed 90%; adding a common completion safeguard reverses residual/maximin stage costs from 136.09/198.25 to 273.63/263.25 on 32 seeds.

Published-rule adaptations. Printed G-ACOL [1] measures the most uncertain remaining configuration. Authors-code commit 33fc4d2 instead selects that target b_u and maximizes its one-report variance reduction,

$$\frac{(b_u^T V a_e)^2}{\sigma_e^2 + a_e^T V a_e}, \quad V = (M + .1I)^{-1},$$

where M is accumulated weighted information. Neither rule normalizes by price; regularization affects allocation only. We adapt noise, history, stage access, batching, and the common v10 `JointVerifier`, without transferring published guarantees.

The frozen comparison uses 64 paired sequences (seeds 57000000–57000063), goals $24 \rightarrow 12.6 \rightarrow 9.5 \rightarrow 24$ ms, $\epsilon = \delta = .05$, and retained initial counts

$$(1024, 1024, 1024, 1024, 0, 0, 1024, 0, 0, 0)$$

in saved probe order. All retain the original window/caps. Batches contain at most eight reports; residual replans within 512 with inflation 1.1. All verify after batches/phase entry; greedy phase endings add checks without observations. Intervals use 10000 complete-sequence paired bootstraps (seed 57009999).

Lookahead is cheaper at 12.6 ms (9.625/9.97625), although residual costs less over every complete sequence. No method uses a stage probe; all loose and 50/64 intermediate goals need no reports. Controller times .124/.225/.218 s include unequal checks and simulation. An unchanged-code check correctly returns arm 8 on eight compatible nine-arm instances (mean 290 reports), without a residual comparator. The original source identifies a separate evidence record set for these checks; it was not supplied. These checks establish neither superiority over unchanged G-ACOL nor Fast Beam Alignment; maximin uses separate tapes.

Table 1: Frozen 64-sequence allocation comparison. Every rule returns all 256 decisions correctly. Savings use each comparator’s mean sequence cost; intervals are paired descriptive 95% bootstrap intervals.

Rule	Mean cost	Saving [95% interval]
Residual	2675.92	—
Code lookahead	2869.31	6.74% [6.43,7.05]%
Maximum variance	2881.81	7.14% [6.83,7.47]%

1.2 Stopping, Calibration, and Continuation Selection

Fixed reserve. The archived Gaussian comparison shares preparation, reserve, confidence, and potential reports across 120 paired attempts: 45 certify initially, 44 after the pilot, 14 during completion, and 17 remain unresolved with infeasible completion designs. For full-quota, block, proportional, and goal-directed execution, the respective mean costs are 4567.93, 1194.59, 1023.80, and 1022.73. The corresponding 14-trial means are 32654.50, 3740.21, 2276.29, and 2267.14. Goal-directed savings are therefore 77.6% overall, 93.1% conditionally, and only .10% against proportional stopping. A separate deterministic 93.5% illustration is not a stochastic estimate. Original raw execution records are missing.

The separate calibration comparison is described only by reference to an unavailable earlier supplement; its population and risk grid are not pooled here.

Continuation selection. Two separate 40-decision studies retain the same resources and mixture verifier ($V_0 = 10I$), using four cases, five seeds, and risks .05, .01. The selector combines directional/balanced/single-probe bases, four discovery reports/probe, at most 100 design iterations, and 512 simulated futures/candidate. Clopper–Pearson/transport allowances are .01 each. The declared holdout replaces 18 quota candidates by six full-quota candidates; the complete checkpoint and tie records were not supplied.

Initial baseline/selector correct counts are 40/32 at costs 3396.68/3534.15; holdout counts are 40/37 at 2881.20/7872.13. No wrong return occurs. Three server-only plans spend 49996 despite 512/512 simulated successes. All nonterminal gap bounds equal one; maximal selected lower bounds are $1.47 \times 10^{-35}/1.68 \times 10^{-159}$. The transport bounds are valid but uninformative. Median/95th-percentile computation is 60.8/154.5 ms for the corrected selector versus 16.5/210.2 ms for baseline; unequal checkpoints confound allocation-cost attribution.

1.3 Audited History and Guarded Acquisition

Paid audits. The earlier Gaussian study crosses physical residuals $b \in \{0, .15, .35\}$, historical shifts $d \in \{0, .35, 1\}$, 80 tapes/cell, and budgets $(B, H) = (5000, 4000), (14000, 9000)$: 1440 attempts/method. Current means are (1, 1), aggregate/stage limits (2.9, 1.9), and the fallback costs one ($\epsilon < 1$). Unit-variance aggregate/stage-2 probes have prices (1, 8), durations (1, 4), 512 historical reports each, shifts $(d, d/2)$, and total/new caps 4000/3488. Auditing acquires 128 reports/probe (cost 1152, duration 640) and admits history when

$$U_e = |\bar{Y}_{e,h} - \bar{Y}_{e,128}| + \sqrt{2(1/512 + 1/128) \log(4/.005)} \leq .9.$$

Audit and four-mask risks are .005 each; the archived verifier adds shift and physical envelopes. Discarded history consumes quota. Fresh/audited pilots are 16/128; adaptive batches of at most 16 maximize predicted certificate-ratio improvement/price, while proportional acquisition targets aggregate/server counts $A/S = \sqrt{8}$. Sufficient plans precede adaptive discovery, with issuance/failure allowance .105 and return risk .01.

Of 1312 post-audit attempts, 37 issue plans; all discard history and complete under the larger budget, reserving/spending 11682.27/60.97 additional units on average. Immediate/discovery correct counts are 128/1273, with two unresolved. Matched fresh pilots reproduce every audited completion. Adaptive and sufficient excess costs are 9.42 and 9.24, with respective paired 95% intervals $[-2.78, 20.96]$ and

Table 2: Paid audits: mean cost and correct returns out of 1440 attempts. All remaining attempts are unresolved; no incorrect return occurs.

Method	Cost	Correct
Fresh adaptive, pilot 16	2002.24	1432
Audited adaptive	2023.39	1438
Audited sufficient	2023.21	1438
Audited proportional	2011.10	1438
Fresh adaptive, matched pilot 128	2013.97	1438

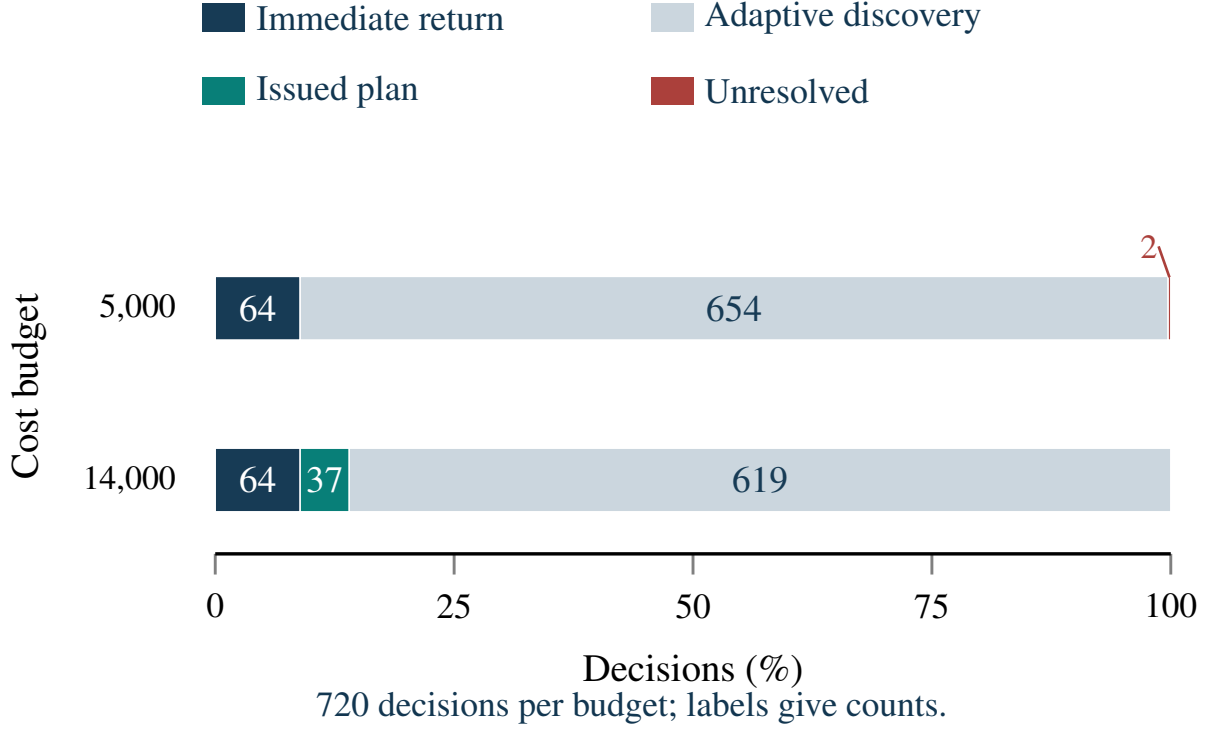


Figure 1: Observed outcomes of the sufficient-plan controller. Each budget uses the same 720 tapes; time allowances are 4000 and 9000 units. All paths include the 1152-unit audit. Successful issued plans are distinguished from immediate verification and adaptive discovery.

$[-2.98, 20.78]$: no net reuse benefit. A separate zero-shift reference's 160 historical completions assume stronger validity information.

Guarded confirmation. Nine methods use 21 declared labels, 32 tapes each, and two budgets (1344 attempts/method); three extension tuples duplicate originals on independent tapes. The primary fixes fresh acquisition actions and uses audited history only in verification/promises. Paired bootstraps preserve labels, tapes, and budgets. Declared targets are a saving lower endpoint above 2% and completion-difference lower endpoint above -1 point.

Primary saving is .138% (95% interval $[\text{.082}, \text{.201}]$ %), below the 2% target. Both primary and guarded fresh complete 575/576 original-grid and 554/768 extension decisions; 192 attempts are boundary/low-margin cases. Only 57 paths save strictly (maximum 304), all respecting fixed-action cost/duration ordering. Relative to fresh repaired, primary costs 1.023% more for one extra completion. Its 4.394% gain over earlier audited controllers changes initial auditing and does not isolate planning. Primary/guarded fresh issue 500/484 promises, and all primary promises complete. Earlier sufficient acquisition issues 39/1152 under a different audit and .105 allowance. Primary mean spent/reserved-additional/actual-additional costs are 874.13/7352.62/347.89; advertised/actual totals are 8226.76/1222.03 (median ratio 6.75). Selected successes and unacquired endpoint replays establish no uniform conditional guarantee. Separate four-

Table 3: Guarded confirmation: all 1344 decisions per method. Remaining decisions are unresolved; none is incorrect.

Method	Mean cost	Correct
Fresh proportional	3021.64	1128
Fresh repaired	2962.29	1128
Fresh slack	2947.11	1128
Guarded fresh	2996.73	1129
Audited, history-guided	2994.15	1128
Audited, balanced	3015.73	1129
Audited, fixed fresh actions	2992.60	1129
Earlier audited repaired	3130.14	1128
Earlier sufficient	3130.14	1128

goal epochs share an evidence/cap ledger and risk .03/4 across thresholds $(5.8, 4.8) \rightarrow (2.9, 1.9) \rightarrow (1.4, .9) \rightarrow (5.8, 4.8)$. Both primary/guarded fresh complete 563/576 epochs, costing 2196.56/2210.54 (saving .633%, interval [.319, 1.022]%). Primary can cost more at goal three; final relaxed goals need no reports. History-guided/balanced variants cost 2184.72/2379.14 with 563/562 completions.

The separate nine-pair confirmation uses 32 new four-goal sequences/interface. Residual reaches estimated alternative distance $1.10\beta^2$ at minimum cost. Custom information-maximin maximizes minimum distance using an added-cost planning allowance of current-goal expenditure plus $\sum_e c_e$ (64 iterations, tolerance .005). This allowance is not a hard budget; the shared executor enforces original count/time limits, unlike resource-aware planning in the acquisition-design section of the theoretical supplement. Residual/scalar costs are 2661.84/3308.72 (19.55%, interval 18.46–20.40%); residual/custom-maximin costs are 2739.90/2806.12 (2.36%, interval 2.13–2.57%). All 512 decisions are correct; 22/32 intermediate and all loose requests need no reports. The development/final attempts, detailed audit records, and vendored comparison source cited by the original manuscript were not supplied. The two-instrument controller is not evaluated on nine pairs.

2 System Boundaries and Evidence Scope

2.1 Instrumentation, Priors, and Agent Operation

The four-configuration model has $\theta = (r_0, r_1, w_{00}, w_{01}, w_{10}, w_{11})$ and $D_{ij} = r_i + w_{ij}$, including nominal stage terms. Aggregate rank is 4, increasing to 5/6 with one/two radio counters. Charges are (1.6, 2.4, 2.6, 3.4), tolerance .05, and total/radio/server limits 24/3.8/8.5 ms. Reports average 64 independent finite-buffer FCFS pre-admission workload draws, including rejected jobs: the target is diagnostic mean, not offered-job latency. Studies A–C start with 256 reports/aggregate (sunk 1920), batches 32, and caps 4096. Prefix radii are

$$h_e(n) = \sqrt{(2\sigma_e^2/n) \log(2 \cdot 6 \cdot 4096/\delta_{\text{on}})}.$$

We intersect earlier intervals, priors, and admitted contrasts without resetting goal risk; empty intersections return model conflict. Exact-rational LP-witness audits check algebra separately from coverage.

A: common allocation. Both verifiers use smallest-count acquisition, tied fast radio/slow radio/aggregates, truncated at full batches for deadlines 150/2000. Thirty-two seeds pair states shifted by +2 ms radio/−2 ms server, preserving aggregates/nonnegativity but changing the valid answer (configuration 3/infeasibility). Aggregate-only correct completion is at most δ ; nonabstention may reach 2δ . This construction changes server nominal terms, not merely the FCFS random seed. Joint exclusion uses total delay above $3.8 + 8.5$ without identifying the violated stage; selected-action feasibility still needs stage evidence. At 150, unresolved aggregate-only/single-counter componentwise costs are 169.92/768. Wilson 95% intervals for 32/32 and 0/32 are [89.3, 100]/[0, 10.7]%; two deadlines locate no transition.

Prior correction. The generator’s valid radio minima (3.88845, 1.54168) ms were omitted originally. Replay strengthens these equally, retaining tapes, bands, and goals, and adds no invalid server bounds.

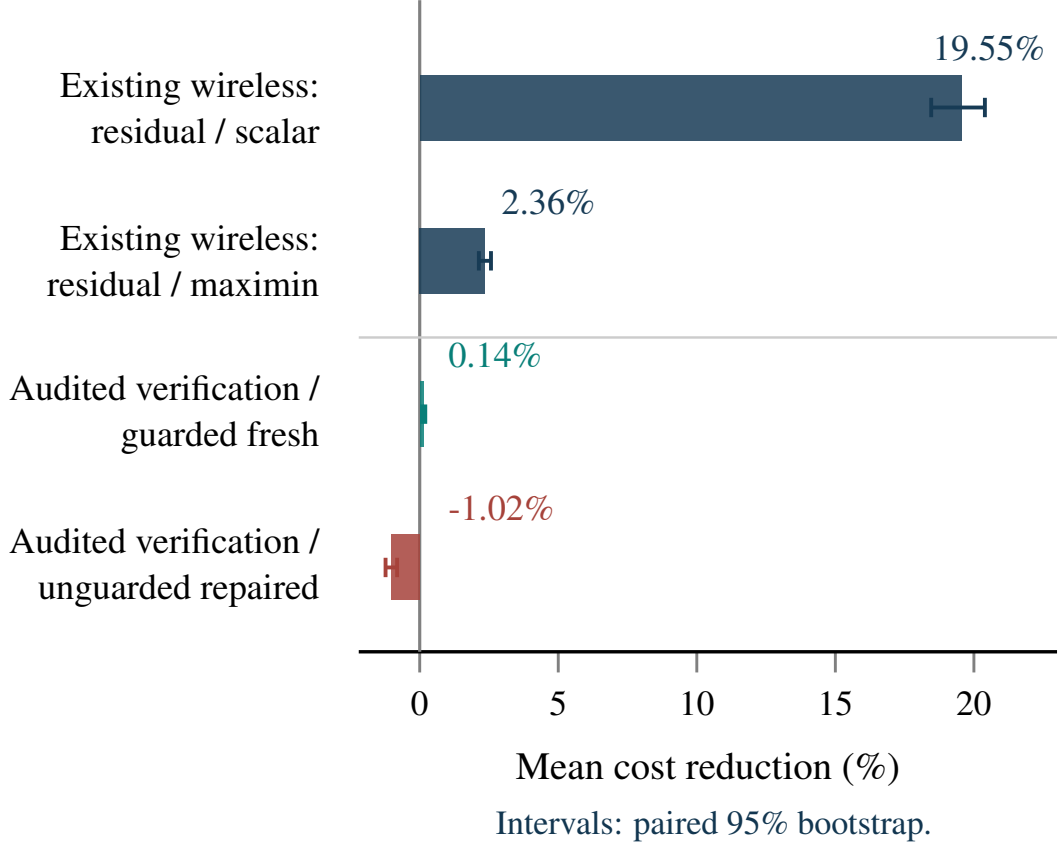


Figure 2: Separate frozen comparisons. The upper rows confirm existing nine-configuration wireless controllers on 32 complete four-goal sequences; the lower rows evaluate the new audited-verification variant on 1344 two-instrument decisions. Positive values denote lower all-attempt cost. The last comparison has one additional completion for the audited variant. Intervals preserve the paired experimental units; the populations and controller interfaces are not pooled.

Both verifiers then complete 32/32 at cost 384/duration 64 in each state: the one-counter advantage disappears. Below, original practical curves explicitly retain the broader zero-lower-bound prior. The full prior-sensitivity replay records were not supplied.

B: interactions. Under radio-dependent server loading, flexible and calibrated-contrast models certify infeasibility from history in all 16 trials; nominal-equality modeling gives 16 conflicts. A constructed common server shift preserves contrasts. All 32 calibration sets cover, with no demonstrated acquisition-cost benefit or physical-transfer guarantee. The supplied text does not specify the separate calibration expenditure, risk split, or full interaction parameters.

C: common verifier. Coverage/custom-maximin/residual share flexible priors, all instruments, 16 seeds/coupling, and duration 2000. Common LP repair produces regularized centers; plans cache at most 256 reports and maximin uses 12 iterations. At zero coupling all complete at costs 384/454.08/524.16 and controller times .038/.076/.143 s. At .24, history completes all at zero new cost. Fixed batches give degenerate paired intervals, not population certainty. This adverse comparison is separate from nine-pair gains.

D: agent loop. One language-model agent accesses public goals/evidence through proposal, acquisition, verification, and authorization tools. Isolation is procedural; the exact model version is unavailable. Loose/joint/loose goals share history; an announced change quarantines it before the final joint goal. Each epoch uses risk .025 and cost/duration caps 2000. Agent and deterministic coverage return 0, 3, 0, then infeasibility. Their totals are 23/45 actions, costs 2133.76/2124.80, and durations 729.44/735.20; the agent acquires 448 reports, none at loose goals. One sequence proves no LLM advantage. The cited attempts,

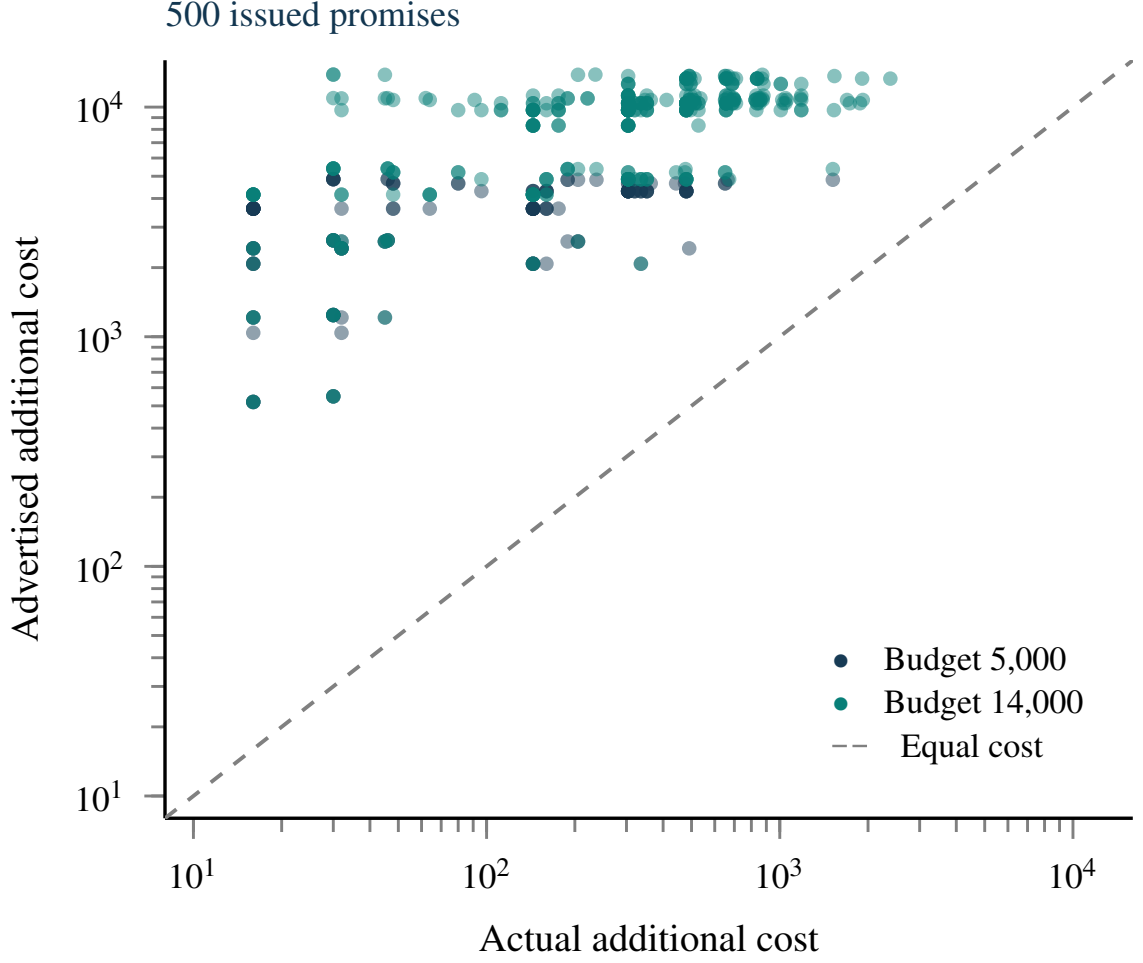


Figure 3: Advertised and actual additional acquisition cost for the primary controller’s 500 issued promises. Both axes are logarithmic. All promises complete in these trials, but reservations are conservative. These selected-promise diagnostics complement the all-attempt comparison in Table 3; they do not establish a conditional probability guarantee.

transcripts, 16 guard checks, and certificate/resource replay files were not supplied.

2.2 Completion Frontiers and Practical Curves

The scalar Gaussian benchmark has cheap-stage means $(6 + \vartheta, 6 - \vartheta)$, limits $(6, 7, 13)$, and a costlier safe backup. For $0 < |\vartheta| < .8$, negative/positive margins uniquely favor cheap/backup. Centered aggregate/radio/server means are $0, \vartheta, -\vartheta$. Radio/server costs are $(1, 6)$, durations $(1, 4)$, and accurate-server variances $(1, 1/9)$, reversed for accurate radio; history is empty and cost/count caps are 512. A public-parameter dynamic program maximizes precision $A_{\max}(H)$ for integer $H \leq 512$. Theorem 1 gives the exact frontier and attaining endpoint test. At zero precision, operational policies abstain although randomization permits 5% correct completion.

At margin .1, accurate-server precision caps at 767 and completion at .869626. Sequential execution orders the same design by precision/duration, spending one-sided risk $.01/[j(j+1)]$ at report $j < N$ and .04 at the endpoint (single-report risk .05). Each deadline defines a separate valid policy.

Evaluation crosses margins $\pm.1, \pm.2, \pm.4$ and deadlines 0, 8, 32, 128, 512: 4096 seeds in each of 480 cells, totaling 1,966,080 correlated outcomes. Maximum endpoint discrepancy across 144 positive-

Table 4: Original zero-lower-bound comparison: correct count / mean new cost in each matched state at duration 2000. Completion counts also hold at 150.

Access	Joint	Componentwise
Aggregate only	0/32;2449.92	0/32;2449.92
Aggregate+fast radio	32/32;384	0/32;6745.92
All six	32/32;384	32/32;768

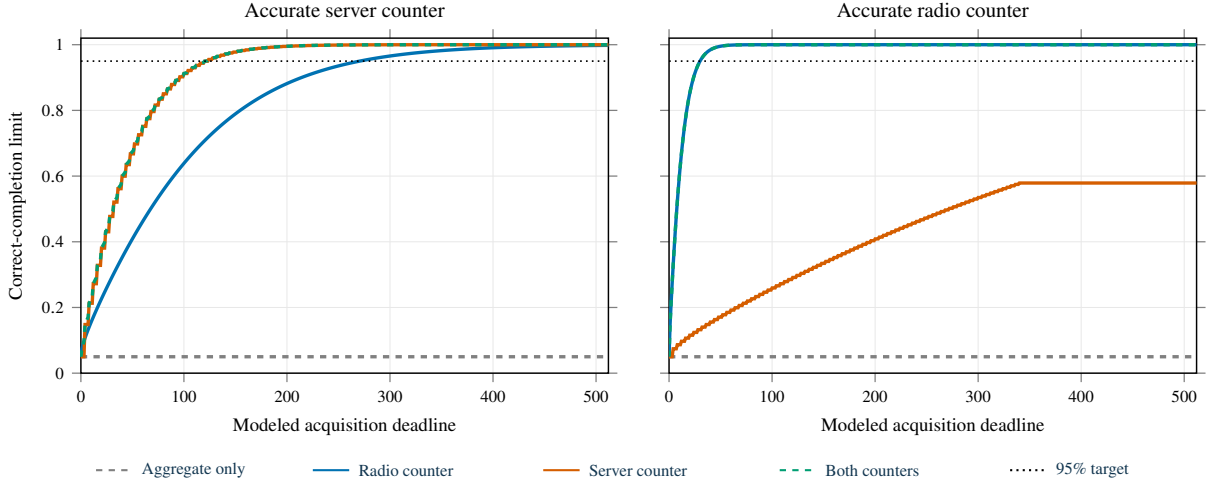


Figure 4: Exact correct-completion limits at absolute stage margin 0.2, wrong-return risk 0.05, and hard cost cap 512. Lines give the exact reference. Counter precision changes which instrumentation can meet a deadline. The aggregate-only line is a randomized 5% upper bound; our operational policies always abstain in that case. Radio and both-counter curves coincide in the right panel. Durations are modeled acquisition units.

information cells is 1.545 points (standardized 2.628). At preselected $\vartheta = .2$, both counters/accurate server/ $H = 128$, endpoint versus sequential returns 3919/3877 correct, 0/2 wrong, and 177/217 unresolved, costing 192/164.883. Other endpoints also trade lower cost for missed or incorrect returns; complete outcome records were not supplied. Early stopping does not improve cost and completion uniformly.

Practical policies. The bounded six-mean/zero-lower-bound model crosses both radio-row orientations, radio limits 3.8/3.1, history 0/256, and all probes versus aggregates plus fixed counter 5. The counter is useful in only one orientation. Caps/risk/batches are 4096/.05/4; history costs 1920. Coverage selects smallest absolute count with seeded random tie order. Goal-directed acquisition targets the cheapest unexcluded configuration and maximizes unresolved directional uncertainty reduction/duration; regularization guides allocation only. One sequence/policy is truncated before unaffordable batches at $H = 0, 16, 32, 64, 128, 256, 512$. Sixteen paired seeds yield 512 traces/3584 deadline views; Wilson and 10000 seed-cluster bootstrap intervals preserve paired orientations/deadlines. These curves are policy-specific.

The wrong-counter obstruction $v = (0, 2, 0, 0, -2, -2)$ preserves aggregate and observed slow-radio laws but changes configuration 3 to infeasibility, respecting the stated prior/noise model. Correct completion is therefore at most δ for every deadline; known server-row equality removes that alternative. Unresolved all-instrument prefixes prove no such impossibility. The original source references additional protocols and design/tape/stopping/certificate audits that were not supplied. The 604 approved deadline views contain no error but are correlated.

Table 5: Practical results at deadline 512. Ordered pairs correspond to radio limits (3.1, 3.8) ms; each correct count is out of 16 seeds. C: coverage; G: goal-directed. Both orientations agree with all probes. Costs include unresolved attempts; unequal completion prevents equivalent-service cost ratios. History adds 1920 sunk units.

Orientation	History/probe	Access	C correct	G correct	C cost	G cost
Both	0	All	(0, 0)	(16, 16)	(1615.10, 1615.10)	(1628.75, 1004.89)
Both	256	All	(15, 16)	(16, 16)	(2517.00, 564.00)	(1275.00, 297.00)
0	0	Fixed 5	(0, 0)	(16, 16)	(1248.00, 1248.00)	(1564.69, 974.63)
0	256	Fixed 5	(16, 16)	(16, 16)	(1275.00, 297.00)	(1275.00, 297.00)
1	0	Fixed 5	(0, 0)	(0, 0)	(1248.00, 1248.00)	(880.04, 918.77)
1	256	Fixed 5	(0, 0)	(0, 0)	(3072.00, 3072.00)	(832.20, 832.20)

Table 6: Exact first modeled deadline for 95% correct completion, wrong-return risk .05, and cost cap 512. Dashes remain unattainable with more time. Both signs agree.

Catalogue	Counter	$ \vartheta = .1$	.2	.4
Accurate server	Radio	—	271	68
	Server	—	124	32
	Both	—	121	32
Accurate radio	Radio	121	31	8
	Server	—	—	272
	Both	121	31	8

2.3 Physical Evidence and Reproduction Limits

No experiment uses a live wireless-edge link. Additive-model holdout RMSE rises from .166 ms under shared load to 2.066/4.449 ms with server interactions (noiseless 2.047/4.386): sampling cannot remove structural error. Historical EdgeDroid [2] and localhost summaries lack the raw observations, harnesses, splits, or timestamp logs needed for reproduction. Missing latencies are not imputed; observed residuals do not validate unseen stage delays.

The original manuscript reports 18 synthetic integrity cases that verify only the record schema; their source files were not supplied.

3 Simulation-Based Continuation Selection

The following construction preserves the full simulation-based selector guarantee from the original supplementary source. Probe e has Gaussian mean $a_e^\top \theta$, variance σ_e^2 , and one-report information matrix $G_e = a_e a_e^\top / \sigma_e^2$. The set Q_{int} denotes the original legal integer fresh-count vectors under the total count, time, and cost caps; discovery counts remain charged. For $M \succeq 0$, write $\|v\|_M^2 = v^\top M v$, and use $P^\dagger$ for the Moore–Penrose inverse. The common verifier has a simultaneous unconditional wrong-return risk at most δ under every admissible acquisition rule. The notation $P^*(\delta, B)$ denotes the paper’s unrestricted correct-completion frontier under this risk and resource bound. These are the assumptions under which the construction below applies.

3.1 An Implementable Continuation Guarantee

At a legal discovery stopping time, let $\mathcal{H}$ contain retained/discovery reports and d the discovery counts. Completed returns are absorbing; otherwise K_c executable candidates $\pi_k(\mathcal{H})$ share the original verifier and remaining set $Q_{\mathcal{H}} = \{n \in \mathbb{Z}_+^E : d + n \in Q_{\text{int}}\}$. The fixed-goal Gaussian subclass has a unique answer at every admitted regular coefficient.

Choose a center $\vartheta_0 = \tilde{\vartheta}(\mathcal{H})$ and an anytime set $C(\mathcal{H})$ of coverage $1 - \delta_{\text{est}}$. For a set $Q_{\mathcal{H},k}$ containing

all possible candidate counts, put

$$\rho_k = \sup_{v \in C(\mathcal{H})} \max_{n \in Q_{\mathcal{H},k}} \|v - \vartheta_0\|_{M_{\text{new}}(n)}, \quad M_{\text{new}}(n) = \sum_e n_e G_e.$$

For $C = \{v : \|v - \vartheta_0\|_P^2 \leq 2b\}$, $P \succeq 0$, $b > 0$, and componentwise quotas n_k , a valid radius is $\rho_k^2 = 2b\lambda_{\max}(P^{\dagger/2} M_{\text{new}}(n_k) P^{\dagger/2})$ when $\text{range}(M_{\text{new}}(n_k)) \subseteq \text{range}(P)$; otherwise use infinity.

Let q_v^k be the conditional probability of any verified return when future reports use coefficient v , holding actual history fixed. Prespecify candidates and L_{MC} independent simulations per candidate under ϑ_0 . Simultaneous conditional $1 - \delta_{\text{MC}}$ intervals $[l_k^{\text{MC}}, u_k^{\text{MC}}]$ follow from Hoeffding half-width $\sqrt{\log(2K_c/\delta_{\text{MC}})/(2L_{\text{MC}})}$, or Clopper–Pearson tails $\delta_{\text{MC}}/(2K_c)$. Candidates may share randomness. Define $T_{\pm}(q, r) = \Phi(\Phi^{-1}(q) \pm r)$, with continuous endpoints, and

$$L_k = T_-(l_k^{\text{MC}}, \rho_k), \quad U_k = T_+(u_k^{\text{MC}}, \rho_k). \quad (3)$$

Use $[0, 1]$ if $\rho_k = \infty$. Execute a maximizer $\hat{k}$ of L_k and put $\Delta = \max_k U_k - L_{\hat{k}}$; an approximate selector uses its actual lower bound.

Theorem 2 (Finite-confidence continuation intervals). *With probability at least $1 - \delta_{\text{est}} - \delta_{\text{MC}}$ over history, discovery, and simulation randomness, simultaneously*

$$L_k \leq q_{\theta}^k \leq U_k \quad \text{for all } k, \quad \max_k q_{\theta}^k - q_{\theta}^{\hat{k}} \leq \Delta.$$

The composed controller is uniformly δ -safe. For conditional correct-completion probabilities $p_{\theta}^k(\mathcal{H})$, define the same-discovery oracle for discovery strategy D , $P_{D,K_c}^(\theta) = \mathbb{E}_{\theta} \max_k p_{\theta}^k(\mathcal{H})$. Then*

$$\begin{aligned} \Pr_{\theta} \{\text{correct return}\} &\geq \mathbb{E}_{\theta} L_{\hat{k}} - \delta_{\text{est}} - \delta_{\text{MC}} - \delta, \\ \Pr_{\theta} \{\text{correct return}\} &\geq P_{D,K_c}^*(\theta) - \mathbb{E}_{\theta} \Delta \\ &\quad - \delta_{\text{est}} - \delta_{\text{MC}} - \delta. \end{aligned} \quad (4)$$

Proof. Condition on $\mathcal{H}, k$. On $\theta \in C(\mathcal{H})$, every continuation has Gaussian information separating θ, ϑ_0 at most ρ_k^2 . The Gaussian information-time comparison proved in the associated paper and Neyman–Pearson applied to return and its complement give

$$T_-(q_{\vartheta_0}^k, \rho_k) \leq q_{\theta}^k \leq T_+(q_{\vartheta_0}^k, \rho_k).$$

Actual history stays fixed; the selected center is not assigned a fixed Gaussian law. Combine monotonicity with simultaneous simulation coverage, then subtract endpoints for Δ . No independence between confidence events is required.

Conditional on history and selection randomness, physical future reports are independent of simulations, so the composed return probability is $\mathbb{E}_{\theta} q_{\theta}^{\hat{k}}$. Integrating the interval/gap bounds costs at most $\delta_{\text{est}} + \delta_{\text{MC}}$. Since $q_{\theta}^k \geq p_{\theta}^k$, subtract the common verifier’s unconditional wrong-return risk δ once to obtain (4). Discovery returns remain absorbing. $\square$

The guarantee provides neither conditional safety after selected history nor an impossibility test when L_k is small. Larger quotas may enlarge ρ_k . Selection costs $O(K_c L_{\text{MC}} N T_{\text{pl}})$ for at most N simulated reports and per-report computation T_{pl} ; charge this when the deadline includes computation.

Comparison to unrestricted P^* additionally subtracts the restriction gap

$$G_{D,K_c,\mathcal{V}}(\theta) = P^*(\delta, B) - P_{D,K_c}^*(\theta) \geq 0.$$

The finite-class oracle shares discovery, resources, and verifier, with θ a pointwise design constant, not implemented input. Hard budget B makes its mean-cost cap redundant; a smaller expected-cost ceiling requires a separate feasibility certificate. The experiments do not bound this restriction gap. The optional quota-selection variants referenced by the original source were not supplied.

3.2 Numerical Adaptive-Instrument Reference

The separate finite-resource adaptive-separation example uses a near stage with absolute margin between $d_{\min}$ and $d_{\max}$ and another stage in $[-D_{\max}, -D]$. Historical observations have fixed stage sums and do not reveal the relevant branch. We retain the full numerical specification:

$$(d_{\min}, d_{\max}, D, D_{\max}) = (0.1, 0.25, 1, 2), \quad (\sigma_D^2, \sigma_1^2, \sigma_2^2) = (25, 1, 1).$$

Coarse reports measure both stages, while a fine report measures its selected stage. Their durations are $(1, 4, 4)$, prices are $(1, 8, 8)$, and report caps are $(300, 400, 400)$. We use $(\delta, \delta_D, \delta_C) = (0.05, 0.01, 0.04)$, deadline $H = 1500$, and budget $B = 3000$. The adaptive rule takes 224 coarse reports, selects the larger coarse mean, and acquires 307 fine reports of that stage. Its duration is $224 + 4 \cdot 307 = 1452$ and its cost is $224 + 8 \cdot 307 = 2680$. The fixed-plan lower bounds in the theoretical supplement, including randomized plans, are 2164.43 duration units and 4328.87 cost units. This is the stated Gaussian numerical specialization under a separated-branch promise, not a replay of a physical deployment or evidence of general adaptive optimality.

References

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